# A Representation-Centred Evaluation of Latent Diffusion for Afrikaans Text Generation

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## Abstract

Latent diffusion generates text by decoding latents from an autoencoder, so every sample depends on that autoencoder. We adapt the pipeline to Afrikaans with limited in-domain data (NCHLT), using an mT5-base encoder–decoder with a Perceiver bottleneck and two diffusion denoisers. The original autoencoder reproduces only 11 of 81 validation passages exactly, yet its diffusion samples score well: test mGPT perplexity is below 1.20 times that of an autoregressive (AR) baseline, the success criterion set before the experiments. Controls show why this is weak evidence: random latents through the same decoder score almost as well while rarely being Afrikaans, and a Gaussian fitted to real latents comes close to the denoiser. A full-training recipe with a longer latent and extra Afrikaans data raises exact reconstruction of reserve passages from 19 to 73 of 103, but diffusion on these latents scores worse: inconclusive against the AR rule on raw perplexity for the larger denoiser, and failing it after length standardisation. Reconstruction, language identity and controls that bypass the denoiser should be reported with generation scores.

## 1 Introduction

A government notice states that the Departement van Maatskaplike Dienste en Armoedeverligting (Social Services and Poverty Alleviation) offers 44 bursaries a year, 20 of them for social development. Encoded into a latent and decoded again by an Afrikaans text autoencoder, the passage keeps 44, loses 20, and the department becomes "Maatskaplike Ontwikkeling" (Social Development). Latent text diffusion generates through such an autoencoder: a denoiser produces a latent, and a decoder turns it into words (Lovelace et al., 2023). In unconditional generation there is no source passage to compare against, so scores on generated text cannot show whether the autoencoder preserves content. Reconstruction does not measure the quality of generated text; it tests whether the latent space represents passages faithfully, which a latent generator relies on to express specific content such as names and numbers.

Latent text diffusion has been evaluated mainly on English, by scoring generated text (Li et al., 2022; Gong et al., 2023; Lovelace et al., 2023). Lovelace et al. (2023) report near-perfect reconstruction for English and, with a jointly fine-tuned mT5, for German, both with large training corpora. We found no study that tests whether such an autoencoder still preserves passages when adapted with limited in-domain data, or what generation scores show when it does not. Afrikaans is an informative case: the NCHLT (National Centre for Human Language Technology) corpus offers licensed government-domain text (Eiselen and Puttkammer, 2014) but only about 1.2 million training words, while Afrikaans is covered by a pretrained encoder–decoder, mT5 (Xue et al., 2021), and an external scorer, mGPT (Shliazhko et al., 2024). We ask:

1. **RQ1:** What constraints does a reduced English reproduction of the pipeline identify before the Afrikaans results are interpreted?
2. **RQ2:** Do autoencoders adapted to NCHLT Afrikaans preserve enough of a passage for faithful reconstruction?
3. **RQ3:** Does Afrikaans latent diffusion compare credibly with an autoregressive (AR) baseline under mGPT perplexity, diversity and repetition diagnostics, and tokenisation fertility?

Before any experiments, we wrote a plan (supplied in the repository) that hypothesised the

adapted pipeline would learn a representation sufficient for reconstruction-quality generation. Under its fixed criteria, the system is credible if its reconstructions preserve content, its mGPT perplexity is at most 1.20 times that of the AR baseline, and its outputs show no severe repetition or loss of diversity. With no published Afrikaans threshold, the 20% tolerance is a pre-set rule separating approximate competitiveness from clear degradation; the ratio is always reported. The plan called for a matched AR baseline; ours matches corpus, splits and tokeniser, but is trained from scratch with one seed and sampled with nucleus sampling, whereas the diffusion system decodes with a pretrained decoder and beam search.

We answer these questions mainly as a diagnostic study. Our central finding is that favourable generation scores and unfaithful reconstruction coexist in the same system, so raw perplexity did not establish reconstruction fidelity. We contribute (i) a reconstruction analysis that separates use of the latent from preservation of the passage; (ii) evidence, from random and fitted-Gaussian latent controls, language identification and a second scorer, that raw generation scores do not track this failure and can favour non-Afrikaans output; (iii) a follow-up autoencoder screen with rules fixed in advance, showing that a revised training recipe substantially improves reconstruction at the original latent size, and that on its latents the larger denoiser beats simple latent priors by a wide margin but no longer clearly meets the AR rule; and (iv) the same checks applied to a Swahili run (Appendix B).

Section 2 explains the methods and metrics, §§3–5 cover related work, data, models and evaluation; §6 reports the results, §7 answers the research questions, §8 gives limitations and departures from the plan, and §9 concludes. The appendices give the model and estimators, the Swahili checks and the evaluation sets.

## 2 Background

**Diffusion models.** A diffusion model learns to reverse a fixed noising process. Training corrupts a data point $x$ into $z_t = \sqrt{\gamma(t)}\,x + \sqrt{1-\gamma(t)}\,\epsilon$ with Gaussian noise $\epsilon$, where the signal level $\gamma(t)$ falls as $t$ goes from 0 to 1 until $z_t$ is close to pure noise, and trains a network, the denoiser, to recover the clean signal, or an equivalent target such as the noise or the velocity, from $z_t$ and $t$ (Ho et al., 2020; Salimans and Ho, 2022). Generation starts from noise and applies the denoiser repeatedly (here for 250 steps), each step removing a little noise. Every step sees the whole sequence, so diffusion produces all positions together, whereas an autoregressive (AR) model predicts one token at a time from the tokens before it.

**Latent diffusion for text.** Text is discrete, so continuous diffusion needs a continuous representation. Latent diffusion first trains an autoencoder: an encoder maps a passage to a compact continuous latent, and a decoder maps the latent back to text (Rombach et al., 2022; Lovelace et al., 2023). The diffusion model then learns the distribution of latents, and each generated latent is turned into text by the decoder. Generation can be no better than the decoder's ability to express a latent as the intended text. Reconstruction, encoding a real passage and decoding it again, measures how much of the passage the latent preserves.

**Perceiver bottleneck.** The mT5 encoder produces one vector per input token, so its output length varies with the passage. A Perceiver (Jaegle et al., 2021) uses a fixed set of learned query vectors that attend to these token vectors through cross-attention, giving a latent of fixed size (here 32 slots of 64 dimensions) whatever the passage length. A second small network maps the latent back to vectors that the mT5 decoder attends to.

**Perplexity.** For a scoring language model $q$ and a set of texts with $N$ tokens in total, perplexity is $\exp\big(-\frac{1}{N}\sum_i \log q(y_i \mid y_{<i})\big)$, the exponentiated mean negative log-likelihood per token; lower values mean the scorer finds the text more predictable. We use an external scorer, mGPT (Shliazhko et al., 2024), because latent diffusion gives no token likelihood comparable with an AR model's. Perplexity rewards predictable text, so it can favour short, repetitive or generic output and does not measure meaning (Holtzman et al., 2020); we therefore report it with length standardisation, language identification and diversity statistics.

**chrF and exact match.** chrF (Popović, 2015) compares a hypothesis with a reference through the F-score of their character $n$-grams (by default up to six characters, with recall weighted above precision), on a scale of 0 to 100. It gives partial credit for near-matches such as a changed inflection. Exact match is stricter: the reconstruction must equal the source. Neither checks meaning, and one

changed numeral costs little chrF but can change what a passage says, so we also count numerals.

**Language identification and fertility.** A language identifier assigns each text its most likely language; we use GlotLID (Kargaran et al., 2023). Tokenisation fertility is the mean number of subword tokens per word. Higher fertility splits words into more pieces, lengthens sequences, and has been linked to lower accuracy in multilingual models (Rust et al., 2021; Lundin et al., 2025).

## 3 Related work

Diffusion models generate by reversing a gradual noising process (Ho et al., 2020; Nichol and Dhariwal, 2021). For text, tokens have been handled by diffusing over discrete states (Austin et al., 2021), over continuous embeddings (Li et al., 2022; Gong et al., 2023; Strudel et al., 2022; Dieleman et al., 2022), or with likelihood-trained continuous models (Gulrajani and Hashimoto, 2023). Latent diffusion instead denoises in an autoencoder's latent space (Rombach et al., 2022); Lovelace et al. (2023) apply this to language by compressing a pretrained encoder–decoder, and Zhang et al. (2023) extend it to paragraphs. A latent model can generate only what its decoder can express: strong decoders may ignore their latent (Bowman et al., 2016; Li et al., 2020), and bottleneck autoencoders trade compression against reconstruction (Montero et al., 2021).

Two recent studies also ask where quality is lost. Gavrilov et al. (2026) separate a codec's reconstruction loss from its latent generator's and find most loss before generation begins; Jiang et al. (2026) find that alignment of the latent space with a pretrained language model matters more for generation than reconstruction fidelity alone. Both use large English corpora; we study an autoencoder adapted with limited in-domain data, with controls showing what external scores reward when its reconstructions are unfaithful.

Open-ended generation has no reference output, so it is usually scored with an external language model and diversity statistics. Probability and quality can disagree (Holtzman et al., 2020; Meister et al., 2020), which motivated distribution-level measures (Pillutla et al., 2021), and any metric needs validating for its use (Reiter, 2018). Resources for African languages remain uneven (Joshi et al., 2020; Nekoto et al., 2020), although models pretrained on modest data can be competitive (Ogueji et al., 2021; Adebara et al., 2023) and Afrikaans-specific encoders exist (Ralethe, 2020). We found no evaluation of diffusion-based generation for Afrikaans.

## 4 Data and method

**Corpus.** We use the clean 2.0 distribution of NCHLT Afrikaans (government-domain text, CC BY 2.5 South Africa) (Eiselen and Puttkammer, 2014). Text is NFC- and whitespace-normalised, contact details are masked, and passages of at most 50 words are formed; passages under three words and exact duplicates are removed, and no language-identification filter is applied. Documents are split with seed 42 into 47,767 training, 5,905 validation and 5,692 test passages (2,005, 251 and 251 documents); no passage is shared across splits, although some 12-word sequences are: 168 of the 4,827 validation passages outside the reserve (64 of 187 documents) share at least one with training. We keep the mT5 tokeniser (250,100 entries, 128-token cap) instead of training a SentencePiece vocabulary (Kudo and Richardson, 2018) as proposed, because the pretrained embeddings depend on it; validation fertility is 1.933 tokens per word, and the cap truncates 95 training passages (0.20%) and 2 of the 4,827 validation passages.

**Autoencoder and diffusion.** Following Lovelace et al. (2023), the mT5-base encoder feeds a three-layer Perceiver (Jaegle et al., 2021) that compresses its states to a latent of 32 slots (vectors) of dimension 64, each rescaled to norm eight, and a reconstruction map passes it to the mT5 decoder. Only the Perceiver components, the final decoder block and the final layer norm are trained, with teacher-forced cross-entropy (CE); we use update 30,500 (effective batch eight) of a run that continued an earlier pilot. The follow-up screen (§6.4) trains every parameter instead. Denoisers are trained to predict the velocity (Salimans and Ho, 2022) under a cosine schedule (Nichol and Dhariwal, 2021), with self-conditioning (Chen et al., 2023); Appendix A gives the training objective, sampler and scoring estimators as implemented. The code is a fork of the official implementation of Lovelace et al. (2023).

**Systems.** The small and large denoisers (width/depth/heads 256/4/4 and 768/12/12; 8.1M and 188M parameters) are trained with seeds 42–46, selecting the checkpoint with the lowest velocity mean squared error of an exponential

moving average (EMA) of the weights on a validation probe. Each run samples latents with 250 denoising diffusion probabilistic model (DDPM) steps (Ho et al., 2020) and decodes them with beam search (four beams, repetition penalty 1.2, no repeated trigrams), producing 1,000 validation and 1,000 test samples. The AR baseline is a four-layer GPT-style model (width 256; 67M parameters, of which 64M, or 95%, are its tied 250,100-entry embedding) trained from scratch on the same splits (Table 1); its checkpoint at update 60,000 is selected on validation CE, and it produces 256 validation samples with nucleus sampling ($p = 0.95$, temperature one). Two latent controls replace the denoiser and are decoded with the same decoder (validation only, 1,000 latents each): isotropic Gaussian latents with each slot rescaled to norm eight, and draws from a Gaussian fitted to the 47,767 training latents (Ledoit–Wolf shrinkage), projected to norm eight per slot.

## 5 Evaluation

**Reconstruction.** On 81 validation passages from 64 documents we report teacher-forced CE and, for free-running output, exact matches against the 128-token model view, mean sentence chrF (Popović, 2015), and numeral recall and precision, counting numerals (`\d+([.,]\d+)*`) as multisets, so dropped, duplicated and invented numerals all count, though not which entity a number belongs to. As a control, each latent is replaced by that of a passage from another document of similar length. Claude Opus 5.5 (Anthropic), shown each source beside its reconstruction and aware of the study aims, also labelled all 81 against three pre-specified content checks (entities and content words preserved, main clause recognisable, nothing unsupported); no Afrikaans reader verified the labels.

**Generation.** All text is scored with mGPT (FP32, 512-token window, stride 256), each output separately after a start token. Perplexity (PPL) is token-weighted over a sample set (§2). We also report mean length, distinct-1 and distinct-2 (unique word $n$-grams over all $n$-grams in a set), the mean share of repeated word 3- and 4-grams within an output, exact training copies and the share of outputs identified as Afrikaans by GlotLID v3 (Kargaran et al., 2023), with a dictionary check against the LibreOffice `af_ZA` Hunspell list. Because mGPT is strongly length-dependent, comparisons are repeated with length-standardised perplexity, which reweights per-band log-loss (1–10, 11–20, 21–35 and 36+ words) to the human length distribution, and under a second scorer, the monolingual Afrikaans Goldfish model (Chang et al., 2024). Goldfish's sources include Glot500, which contains NCHLT corpora, so overlap with our data cannot be ruled out. Evaluation sets differ in split and size; Appendix C lists every set, how it was chosen and when it was first used, and comparisons are made only within a set.

**Uncertainty and data use.** Diffusion values are means over five runs. The large-minus-small difference has a run bootstrap, a Welch interval and an exact permutation test over all 252 divisions of the ten runs; AR-ratio intervals omit AR and autoencoder training variation; reconstruction intervals resample documents. All intervals are exploratory. Checkpoints were selected on validation data, but the test set was exposed during the project (Appendix C). From 23 October test banks were treated as an exposed benchmark with models frozen, so test results are not fresh confirmatory evidence. Appendix B applies the checks unchanged to an earlier Swahili run of the pipeline.

## 6 Results

### 6.1 Reconstruction

With correct latents teacher-forced CE is 1.079; with shuffled latents it is 8.319 (difference 7.240, 95% CI [6.605, 7.933]), and chrF falls from 67.70 to 12.84 (Table 2). The decoder therefore relies on source-specific information in the latent, yet the encode–decode path loses content: only 11 of 81 passages are reconstructed exactly and 21 of 33 numerals survive, while 4 numerals appear that are not in the source, with no input truncated and no output at the length limit. Five further shuffles give chrF 12.6 to 15.4. On 681 passages never used in development, the held-out slice of the extra data (government statements and newspaper text, longer and richer in numbers than NCHLT; median 64 against 48 tokens), the same autoencoder reconstructs only 13 exactly (1.9%), with chrF 65.6, numeral recall 0.30 and precision 0.59. Exact-match failure is not the same as a change of meaning, since a punctuation difference also fails it, but many errors do change content: "regulering" (regulation) becomes "betaling" (payment), and "Regeringsdepartement" (government department) becomes "Regeringskommissie" (commis-

| Condition | Small diffusion | Large diffusion | AR baseline |
|---|---|---|---|
| Pretrained weights | Adapted mT5 | Same adapted mT5 | None (random initialisation) |
| Generator parameters | 8.10M | 187.98M | 67.22M |
| Training seeds | 42–46 | 42–46 | 42 |
| Precision / effective batch | FP32 / up to 64 | FP32 / up to 64 | BF16 / 8 |
| Exposure at selection | 288–306 epochs | 172–180 epochs | 480,000 sampled passages |
| Recorded training time | 2.93–3.13 h | 7.19–7.85 h | 4.52 h |
| Text decoding | Beam 4, constrained | Beam 4, constrained | Nucleus, $p = 0.95$, temperature 1 |

Table 1: Comparison conditions (RQ3). All systems use the same splits and tokeniser. Diffusion parameter counts exclude the shared autoencoder; the AR count includes its tied 250,100-entry embedding, so counts are not whole-system equivalents. Training times cover training and checkpoint selection on an RTX PRO 6000 Blackwell and exclude pretraining and autoencoder training. AR passages are sampled with replacement, so they are not complete epochs.

| Measure | Correct | Shuffled |
|---|---|---|
| Teacher-forced CE | 1.079 | 8.319 |
| Mean sentence chrF | 67.70 | 12.84 |
| Exact reconstructions | 11/81 | 0/81 |
| Numeral recall | 21/33 | 0/33 |
| Numeral precision | 21/25 | 0/25 |
| All content checks (Claude) | 19/81 | – |

Table 2: Reconstruction of 81 validation passages from correct or shuffled latents (RQ2). Numerals are counted as multisets: 33 in the sources, 25 in each arm's outputs (correct latents: recall 0.64, precision 0.84). Content checks: passages passing all three checks in Claude's reading.

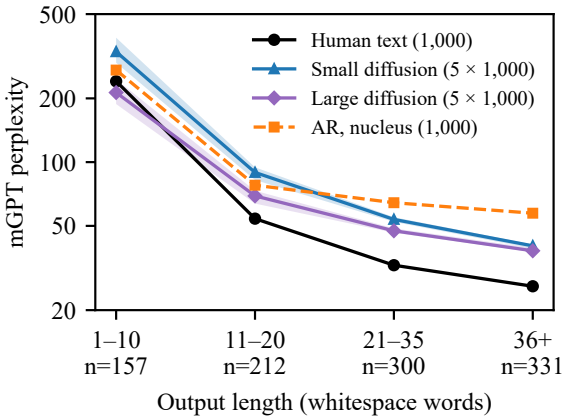



Figure 1: mGPT perplexity by output length on the test split, log scale: 1,000 human passages (counts per band under each label) and five runs of 1,000 outputs per diffusion size (line: mean of run values; shading: range), and 1,000 AR samples.

sion). Only model estimates exist for how often this happens: Claude marked 62 of 81 passages (77%) as failing at least one content check, agreeing with an earlier AI labelling (assistant not recorded) on 74 of 81 ($\kappa = 0.79$). These are unvalidated model labels, not measured error rates; no Afrikaans reader checked them.

### 6.2 Generation

**Denoiser size.** The larger denoiser lowers test perplexity from 51.31 to 48.61 (Table 3; difference −2.704; Welch 95% CI [−4.567, −0.841]; exact permutation $p = 6/252 = 0.024$). Width, depth and exposure change together, so this is the effect of the configuration as a whole. Its outputs copy a training passage more often (79 against 5 of 5,000), though removing copies leaves perplexity almost unchanged (49.60 to 49.56 on validation). On validation, 12 to 20 of its outputs per run are near-copies (at least 90% of word 8-grams found in training), against 0 to 4.

**Diversity and repetition.** Test distinct-1 is 0.223 (small) and 0.234 (large), and repeated word 3-grams make up 0.001% and 0.003% of an output on average (4-grams: 0 and 0.0004%), largely because decoding bans repeated trigrams. On validation, diffusion distinct-1 is 0.32 to 0.36 per run against 0.380 for AR, whose outputs repeat 0.14% of their 3-grams. Neither system shows severe repetition or loss of diversity.

**AR comparison and the decoder alone.** For the test comparison, 1,000 fresh AR samples drawn with the same settings score 79.15 (79.47 length-standardised); the test ratios are 0.648 [0.600, 0.701] and 0.614 [0.571, 0.661], and 0.898 [0.841, 0.958] and 0.752 [0.701, 0.806] after length standardisation, all below 1.20. Random latents decoded by the same decoder, however, score 50.4, 51.7 and 59.5 in three banks of 1,000, against 51.94

| | Validation, 256 samples | | Test, 1,000 samples per run | | | | Validation |
|---|---|---|---|---|---|---|---|
| System | PPL | Ratio to AR | PPL | Mean words | Distinct-2 | Train copies | Afrikaans |
| Human text | 32.82 | – | 33.62 | 27.66 | 0.6796 | 0/1,000 | 98.2% |
| Random latents | 57.60 | 0.7009 | – | – | – | – | 5.5% |
| AR, nucleus | 82.18 | – | 79.15 | 18.14 | 0.6898 | 53/1,000 | 95.3% |
| Small diffusion | 53.07 | 0.6458 | $51.31 \pm 1.49$ | 29.30 | 0.6606 | 5/5,000 | 96.3% |
| Large diffusion | 50.02 | 0.6087 | $48.61 \pm 0.91$ | 26.34 | 0.6903 | 79/5,000 | 97.2% |

Table 3: Generation results (RQ3). Diffusion values are means over five runs ($\pm$ sample SD). Random latents are decoded by the same decoder. Ratio 95% intervals: small [0.5813, 0.7148], large [0.5477, 0.6736]. Random latents were evaluated on validation only. Afrikaans: share of outputs whose top GlotLID label is `afr_Latn`, on validation outputs (human: 4,825 passages, 95% interval 96.5–98.7%).

and 49.60 for the $5 \times 1{,}000$ validation outputs of small and large diffusion, so mGPT perplexity alone does not show what the denoiser adds. Human test text scores 33.62, but the scorer is strongly length-dependent (Table 7). Both diffusion sizes have higher perplexity than human text in every band longer than ten words (Figure 1). Resampling the AR model under 18 temperature, top-$p$ and constraint settings gave no comparator eligible under the pre-set rule of at most 13 exact training copies per 1,000 validation outputs (the diffusion maximum): every setting produced 36 to 112, and the reference setting itself gives 53 on test. Seven settings at temperature 0.8 or 0.9 nonetheless score below both diffusion sizes on length-standardised mGPT (43.4 to 58.7, against 59.6 and 71.3), while copying 68 to 112 training passages per 1,000. A modestly tuned AR model can therefore beat diffusion on this metric, largely by copying more. The AR baseline is thus a bounded from-scratch comparator, not evidence that diffusion outperforms autoregression.

### 6.3 Language, shared phrasing and a second scorer

Most diffusion outputs are identified as Afrikaans, at rates near human text (Table 4); the small configuration falls just below the human interval, mostly in outputs of ten words or fewer. In Claude's reading of 50 unrecognised words per group, 22 from diffusion were malformed against 8 from human text. Diffusion outputs share 8.7% (small) and 11.8% (large) of their word 4-grams with the training split, against 14.4% for held-out human text.

**Tokenisation fertility.** All generated test banks have slightly lower mT5 fertility (Rust et al., 2021; Lundin et al., 2025) than human test text (1.77 to 1.79 tokens per word for both diffusion systems and AR, against 1.86), so generation favours words the tokeniser splits less. On the 681 extra-data passages, fertility tracks reconstruction difficulty for every autoencoder: in the most fragmented third, arm D reconstructs 48% exactly against 66 to 69% in the other two, and the original's chrF falls to 61.3 from about 67.8 (Spearman $\rho$ between fertility and chrF $-0.16$ to $-0.23$). Fertility also rises with passage length ($\rho = 0.30$), so this does not isolate a separate cause.

The random-latent control shows why these checks matter. Its raw mGPT perplexity is within ten points of diffusion's, yet only 5.5% of its outputs are identified as Afrikaans, and 995 of 1,000 in each bank we checked contain literal mT5 sentinel strings (`<extra_id_0>` and similar) that mGPT scores as ordinary text, so its favourable raw

| | | Unrec. | 4-gram | Std. PPL | |
|---|---|---|---|---|---|
| Bank | Afr. | tokens | overlap | mGPT | Gold. |
| Human | 98.2 | 4.3 | 14.4 | – | – |
| Human subset | 97.9 | 4.2 | 15.7 | 41.5 | 64.6 |
| Small diffusion | 96.3 | 5.8 | 8.7 | 71.3 | 194.3 |
| Large diffusion | 97.2 | 5.9 | 11.8 | 59.6 | 148.1 |
| AR, nucleus | 95.3 | 5.3 | 15.8 | 80.5 | 203.7 |
| Random latents | 5.5 | 70.5 | 0.0 | 107.6 | 98.4 |
| Fitted Gaussian | 99.7 | 4.6 | 8.3 | 63.6 | 172.5 |

Table 4: Validation outputs (%, except Std. PPL, the length-standardised perplexity). The 4,825-passage human pool was not scored by mGPT or Goldfish; scorer values are for an 811-passage human subset. Rows: 4,825 human passages; $5\times1{,}000$ per diffusion size; 256 AR; 1,000 latents per control (random and fitted-Gaussian latents). Afr.: share identified as Afrikaans by GlotLID. Gold.: Goldfish. Unrec.: word tokens not in the `af_ZA` dictionary. 4-gram overlap: share of word 4-grams found in the training split. Human intervals (document bootstrap): Afrikaans 96.5–98.7, unrecognised 3.2–5.7, overlap 10.2–19.4.

score partly reflects predictable malformed strings and length weighting. Length-standardised mGPT ranks it last, but Goldfish does not penalise it: its raw Goldfish perplexity is 56.1 against 159.4 and 133.5 for small and large diffusion, and even after standardisation Goldfish ranks it best among the generated outputs, though still behind human text (Table 4). Goldfish is therefore not a validated quality scorer for this use.

The fitted-Gaussian control asks a sharper question: does the denoiser beat a simple model of where real latents lie? Its outputs are identified as Afrikaans (99.7%), with fewer unrecognised words than diffusion output (4.6%), though they are less diverse (distinct-2 0.63 against 0.67 and 0.70). On length-standardised mGPT perplexity the large configuration is lower by 4.0 (95% CI [−7.5, −0.5]) and the small one higher by 7.7 ([3.7, 11.7]). The protocol's primary control, recombined slots from independent passages, produced only 2 outputs of ten words or fewer, too few for this comparison; the fitted-Gaussian contrasts were exploratory and were given this weight after the results were seen. For the named comparisons on the 256 validation AR samples it agrees with mGPT in direction: the large-minus-small difference and the large configuration's advantage over AR are confirmed under both scorers (standardised ratios 0.84 and 0.74 under mGPT, 0.76 and 0.73 under Goldfish), but the small configuration's advantage is not clear under either (mGPT 0.89 [0.76, 1.03]; Goldfish 0.95 [0.78, 1.16]); against 1,000 AR test samples its mGPT ratio is 0.90 [0.84, 0.96] (§6.2). MAUVE (Pillutla et al., 2021) with LaBSE embeddings (Feng et al., 2022) and two local model judges failed their pre-set calibration and are not reported as results.

### 6.4 Interventions and full autoencoder training

Four decoder changes, each with a threshold fixed in advance, were tried on the 81 passages: decoding without the trigram ban, retraining the last two or all twelve decoder blocks, and retraining all twelve with dropout 0.1. None passed its threshold, but missing a threshold is not the absence of an effect. Retraining all twelve raised chrF from 67.70 to 70.05 (paired 95% CI for the change [0.78, 3.96]) and numerals from 21 to 25 of 33, a small gain in one run, and stayed below the 70.70 chrF and 14-exact thresholds. Most of that gain came within 1,320 updates (chrF 69.61); later checks to 26,620 stayed between 67.90 and 70.05. With dropout, chrF stayed between 67.93 and 69.80, and a same-seed run without it tracked the earlier curve within 0.51.

| Arm | Data | Slots | Exact | chrF | Num. |
|---|---|---|---|---|---|
| Original | NCHLT | 32 | 26 | 69.24 | 0.62 |
| A | NCHLT | 32 | 118 | 96.35 | 0.94 |
| B | NCHLT | 64 | 126 | 97.89 | 1.00 |
| C | + extra | 32 | 121 | 96.51 | 0.99 |
| D | + extra | 64 | 134 | 98.20 | 1.00 |
| Reserve, 103 passages | | | | | |
| Original | NCHLT | 32 | 19 | 68.32 | 0.77 |
| D (winner) | + extra | 64 | 73 | 98.70 | 1.00 |

Table 5: Autoencoder screen. Exact: exact reconstructions, out of 194 (comparison panel) or 103 (reserve). Num.: recall of source numerals as a fraction, as in Table 2. + extra: NCHLT plus 511,155 words of other government-domain Afrikaans. Arms A–D train every parameter; the original trains only part of the model. Every arm stopped at its six-hour cap after about 3,500 of 10,000 planned updates.

We then trained four new autoencoders under a protocol fixed before training (Table 5), all following the mT5-base recipe of Lovelace et al. (2023): every parameter trained, AdamW at learning rate $10^{-4}$, effective batch 128, 500 warm-up updates and linear decay over 10,000, with one training seed per arm. The arms cross training data with latent length (32 or 64 slots). The extra data are 15,269 passages (511,155 words) of Gov-ZA cabinet statements (MIT licence) and the Vuk'uzenzele government newspaper (CC BY 4.0) (Lastrucci et al., 2023), processed like NCHLT, with duplicates of NCHLT passages removed and the 3 documents sharing a 12-word sequence with NCHLT validation or test dropped. Validation documents were divided into three disjoint panels: a 663-passage selection panel (whose documents also supply the 81 development passages), a 194-passage comparison panel and 103 reserve passages held back since 22 October. Arms were ranked on the selection panel, and the winner was evaluated once against the original on the reserve. At the original latent size the revised recipe raised exact matches on the comparison panel by 47.4 points (95% CI [40.4, 54.7]); longer latents added 4 to 7 points, and extra data at most 4. These are single-run contrasts. The winner met all three reserve thresholds: 73 of 103 exact against 19 (difference 52.4 points,

CI [42.5, 63.3]), chrF 98.70 against 68.32, and numeral recall 1.00 against 0.77. Exact matches on long reserve passages remain rare (1 of 15), although their chrF is 95.9; whether the other 14 keep their meaning has not been checked by a reader.

### 6.5 Generation on the retrained autoencoder

We trained both denoisers on the winner's 64-slot latents, three seeds each (42–44), for 90,387 updates per run (121 epochs), and evaluated them once on the test banks. Seeds agree closely, but every best checkpoint is the last, so convergence is not established. Generation did not improve: test mGPT perplexity is $95.90 \pm 0.88$ (large) and $288.45 \pm 17.54$ (small), length-standardised 109.7 and 307.2, against 48.61 and 51.31 for the original pipeline; exact training copies are 7 and 1 of 3,000. Against the AR test samples, the large configuration's ratio is 1.21 [1.09, 1.35] raw and 1.38 [1.24, 1.54] length-standardised, and the small configuration's 3.64 [3.03, 4.38]. The comparison is not matched (autoencoder recipe, latent length, training exposure of 121 against 172–306 epochs, and seeds all differ), so it does not show that better reconstruction caused worse scores. Random latents through the new decoder score 512.5 (standardised 671.9) and contain no sentinel strings, and a Gaussian fitted to the new training latents gives standardised perplexity 275 to 278 over three banks of 1,000, against 107.8 to 111.8 for the large runs. Under mGPT the large denoiser thus improves on a fitted prior by a wide margin, against a small one in the original space (59.6 against 63.6); the small one (300 to 314) does not. On the 681 extra-data passages, arms A and D reconstruct 171 and 415 exactly against 13 for the original (D minus original 59.0 points, document-bootstrap CI [55.2, 65.0], from only 17 documents), with chrF 92.2 and 98.2 against 65.6 and numeral recall 0.77 and 0.97 against 0.30. D was trained on other documents from the same two sources, so its figure is in-domain; A's is not. In Claude's reading of five fixed test outputs per size, all read as Afrikaans but none is coherent throughout, and all five small outputs contain non-words.

## 7 Discussion

**Verdict.** *RQ1 is answered: the English control gave an early warning that was not acted on.* A reduced English control (BART on ROCStories, 5,000 updates) reached 93.26% validation token accuracy but reconstructed only 11 of 128 passages exactly (8.6%), in a run that had not converged. High token accuracy with few exact reconstructions is the pattern later found in Afrikaans; at the time we read the control only as evidence that the code runs and learns. It did not exercise the mT5 path. The retrained autoencoders, with the same architecture, make a fault in the shared code unlikely. *RQ2 is answered: no for the original recipe, largely yes for the revised one under automatic measures.* The original autoencoder's reconstructions change names, numerals and content words. A revised recipe raises exact reconstruction of reserve passages from 19 to 73 of 103 and, at the original latent size, reconstructs 171 of 681 unseen out-of-domain passages exactly (chrF 92.2), but readers have not checked that meaning is preserved. *RQ3 is answered: the comparison is not credible.* Against the AR test samples, the original system's ratios are below 1.20 both raw and after length standardisation (upper limits 0.70 and 0.96), but the ratio does not measure what it was meant to: a tuned AR model scores better by copying more, and random latents through the same decoder score almost as well. The plan's definition of credible also requires reconstructions that preserve content, which the original system lacks. For the retrained system, the pre-set raw rule is inconclusive for the large configuration (1.21 [1.09, 1.35]) and clearly failed by the small one (3.64); after length standardisation, a diagnostic not in the original plan, the large one also fails (1.38). *The hypothesis is not supported*: the original system lacked adequate reconstruction, and the system with much better reconstruction did not meet the pre-set ratio, although its diffusion runs had not converged.

**Why scores and fidelity diverge.** Every sample passes through the same pretrained decoder with beam search, and decoder and scorer were both pretrained on multilingual corpora that include Afrikaans. Random latents reach diffusion-level raw perplexity through this decoder, helped by sentinel strings it emits, while reconstructions score 70.67 against 61.76 for their sources, so the decoder does not simply pull text towards high-probability Afrikaans. Raw perplexity therefore cannot separate the denoiser's contribution from the decoder's, and it does not detect output that is not Afrikaans; the fitted-Gaussian control adds that much of what the scorers reward is already reachable by a simple prior over this latent space.

**Under-adaptation or bottleneck?** We call an autoencoder under-adapted when its trainable parameters have not been trained enough to use the latent it has; a bottleneck is a latent too small to hold the passage, and predicts errors that grow with length and gains from a longer latent. On the reserve passages the original reconstructs 16 of 47 short, 3 of 41 medium and 0 of 15 long passages exactly, against 45, 27 and 1 for arm D. At the original latent size the revised recipe added 47 points of exact match, and longer latents only 4 to 7, which argues against a fixed capacity limit as the main explanation. The recipe also changes the batch size (128 against eight), schedule and training history, so the evidence favours a training problem over a bottleneck without showing which change matters.

## 8 Limitations and deviations

The study covers one administrative corpus, one language (Swahili only descriptively) and one task, and training text was not screened with language identification. The diffusion and AR systems differ in pretraining, decoding and compute, so differences are not attributable to diffusion as such. mGPT perplexity is length-dependent, chrF and numeral matching are surface measures, and low repetition is partly enforced by decoding. Seeds are conditional on one autoencoder and one AR run, each screen arm is a single training run, intervention gains are exploratory, and the test set is not untouched. Training rescales the self-conditioning estimate to norm eight per slot but sampling does not (Appendix A); this affects generation, not reconstruction. A paired audit (100 seeds per checkpoint, normalisation matched or not) classed the effect under its pre-set rule as inconclusive for the original denoisers and material for the retrained ones. Matching the normalisation lowered mGPT loss by 0.06 to 0.17 nats per token for the original large denoisers (two of three intervals exclude zero) and by 0.29 to 0.47 for the retrained small ones (all three), with no consistent change for the others. The protocol then required the final-system conclusions to be re-examined under a new protocol; this was not done, so reported results keep the original setting. The audit job was submitted before its protocol was explicitly approved.

Departures from the original plan: the English control was a reduced, autoencoder-only check (BART on ROCStories) rather than the planned OpenWebText reproduction, so generation was not checked in English; the tokeniser is mT5's rather than a new SentencePiece vocabulary; the AR baseline matches corpus, splits and tokeniser only, with one seed; fertility is analysed descriptively rather than against compute; and the planned researcher checklists were not completed. Reconstruction is reported as exact match and chrF, not reconstruction loss. Additions to the original plan are five seeds per configuration, the latent controls, the decoder interventions, the autoencoder screen, diffusion on its winner with follow-up checks, the AR test-split comparison, and the Swahili checks. An early pilot adding a decoder loss to denoiser training is documented but not reported; its success criterion, a blinded human review, was not carried out.

## 9 Conclusion

Afrikaans latent diffusion trained on limited in-domain data outscored a from-scratch autoregressive comparator although its autoencoder does not reproduce its inputs, and latents that bypass the denoiser scored almost as well. Full training substantially improved reconstruction, but diffusion on it did not clearly meet the comparator rule.

Our contribution is to show that, with limited in-domain data, generation scores can look favourable while reconstruction fails, and to supply the checks that expose this: reconstruction analysis, latent controls, language identification and a second scorer, with a pre-set autoencoder screen that found a recipe which substantially reduces the deficit. They should be reported with generation scores in any new language.

Future work should add blinded annotation of reconstruction errors by Afrikaans readers; human calibration of the language-identification and dictionary labels; a converged, matched comparison of the original and retrained pipelines with the self-conditioning normalisation corrected; and a pre-trained AR baseline.

## Statements

**AI use.** AI assistance supported artefact auditing, analysis, sample and reconstruction inspection, drafting and LaTeX preparation. The reported AI readings are by Claude (Anthropic): the content labels and the screen samples by Claude Opus 5.5 on 7 November 2026, and the unrecognised words by an unrecorded version; all labels are in the repository. AI-assisted inspection is not human evalua-

tion, and the author is responsible for the claims. **Ethics.** The corpus is publicly licensed and contact details were masked; no human participants were involved, so no participant-research ethics application was made. **Data and code.** Code, protocols and documentation are in the repository supplied with the submission; `REPRODUCIBILITY.md` at the repository root indexes the training configurations, scorer revisions, length-standardisation weights and band-support rule, and the interval procedures. Each study's folder holds its protocol, code and summary results, with SHA-256 hashes of every output file. Model weights, corpora and full sample banks are not in the repository; the hashes allow regenerated outputs to be checked.

## References


Ife Adebara, AbdelRahim Elmadany, Muhammad Abdul-Mageed, and Alcides Alcoba Inciarte. 2023. SERENGETI: Massively multilingual language models for Africa. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 1498–1537, Toronto, Canada. Association for Computational Linguistics.

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pages 17981–17993.

Samuel R. Bowman, Luke Vilnis, Oriol Vinyals, Andrew M. Dai, Rafal Jozefowicz, and Samy Bengio. 2016. Generating sentences from a continuous space. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning*, pages 10–21, Berlin, Germany. Association for Computational Linguistics.

Tyler A. Chang, Catherine Arnett, Zhuowen Tu, and Benjamin K. Bergen. 2024. Goldfish: Monolingual language models for 350 languages. *arXiv preprint arXiv:2408.10441v3*.

Ting Chen, Ruixiang Zhang, and Geoffrey Hinton. 2023. Analog bits: Generating discrete data using diffusion models with self-conditioning. In *International Conference on Learning Representations*.

Sander Dieleman, Laurent Sartran, Arman Roshannai, Nikolay Savinov, Yaroslav Ganin, Pierre H. Richemond, Arnaud Doucet, Robin Strudel, Chris Dyer, Conor Durkan, Curtis Hawthorne, Rémi Leblond, Will Grathwohl, and Jonas Adler. 2022. Continuous diffusion for categorical data. *arXiv preprint arXiv:2211.15089v3*.

Roald Eiselen and Martin J. Puttkammer. 2014. Developing text resources for ten South African languages. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 3698–3703, Reykjavik, Iceland. European Language Resources Association.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891. Association for Computational Linguistics.

Alexey Gavrilov, Alan-Barsag Gazzaev, and Sergey Muravyov. 2026. Where quality breaks in compressed short-text generation: Staged bottleneck localization. In *Proceedings of the 39th Conference of the Open Innovations Association FRUCT*, pages 69–76. ArXiv:2607.24176v1.

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, and Lingpeng Kong. 2023. DiffuSeq: Sequence to sequence text generation with diffusion models. In *International Conference on Learning Representations*.

Ishaan Gulrajani and Tatsunori B. Hashimoto. 2023. Likelihood-based diffusion language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 16693–16715.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *International Conference on Learning Representations*.

Andrew Jaegle, Felix Gimeno, Andy Brock, Oriol Vinyals, Andrew Zisserman, and Joao Carreira.

2021. Perceiver: General perception with iterative attention. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4651–4664.

Jiaxiu Jiang, Jingjing Ren, Wenbo Li, Bo Wang, Haoze Sun, Yijun Yang, Jianhui Liu, Yanbing Zhang, Shenghe Zheng, Yuan Zhang, Haoyang Huang, Nan Duan, and Wangmeng Zuo. 2026. TextLDM: Language modeling with continuous latent diffusion. *arXiv preprint arXiv:2605.07748v1*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, Online. Association for Computational Linguistics.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schütze. 2023. GlotLID: Language identification for low-resource languages. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6155–6218. Association for Computational Linguistics.

Taku Kudo and John Richardson. 2018. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71, Brussels, Belgium. Association for Computational Linguistics.

Richard Lastrucci, Isheanesu Dzingirai, Jenalea Rajab, Andani Madodonga, Matimba Shingange, Daniel Njini, and Vukosi Marivate. 2023. Preparing the Vuk'uzenzele and ZA-gov-multilingual South African multilingual corpora. In *Proceedings of the Fourth Workshop on Resources for African Indigenous Languages (RAIL 2023)*, pages 18–25, Dubrovnik, Croatia. Association for Computational Linguistics.

Chunyuan Li, Xiang Gao, Yuan Li, Baolin Peng, Xiujun Li, Yizhe Zhang, and Jianfeng Gao. 2020. Optimus: Organizing sentences via pre-trained modeling of a latent space. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4678–4699, Online. Association for Computational Linguistics.

Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori B. Hashimoto. 2022. Diffusion-LM improves controllable text generation. In *Advances in Neural Information Processing Systems*, volume 35, pages 4328–4343.

Justin Lovelace, Varsha Kishore, Chao Wan, Eliot Shekhtman, and Kilian Q. Weinberger. 2023. Latent diffusion for language generation. In *Advances in Neural Information Processing Systems*, volume 36.

Jessica M. Lundin, Ada Zhang, Nihal Karim, Hamza Louzan, Victor Wei, David Adelani, and Cody Carroll. 2025. The token tax: Systematic bias in multilingual tokenization. *arXiv preprint arXiv:2509.05486*.

Clara Meister, Ryan Cotterell, and Tim Vieira. 2020. If beam search is the answer, what was the question? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2173–2185. Association for Computational Linguistics.

Ivan Montero, Nikolaos Pappas, and Noah A. Smith. 2021. Sentence bottleneck autoencoders from transformer language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1822–1831. Association for Computational Linguistics.

Wilhelmina Nekoto, Vukosi Marivate, Tshinondiwa Matsila, Timi Fasubaa, Taiwo Fagbohungbe, Solomon Oluwole Akinola, Shamsuddeen Muhammad, Salomon Kabongo Kabenamualu, Salomey Osei, Freshia Sackey, Rubungo Andre Niyongabo, Ricky Macharm, Perez Ogayo, Orevaoghene Ahia, Musie Meressa Berhe, Mofetoluwa Adeyemi, Masabata Mokgesi-Selinga, Lawrence Okegbemi, Laura Martinus, Kolawole Tajudeen, Kevin Degila, Kelechi Ogueji, Kathleen Siminyu, Julia Kreutzer, Jason Webster, Jamiil Toure Ali, Jade Abbott, Iroro Orife, Ignatius Ezeani, Idris Abdulkadir Dangana, Herman Kamper, Hady Elsahar, Goodness Duru, Ghollah Kioko, Murhabazi Espoir, Elan van Biljon, Daniel Whitenack, Christopher Onyefuluchi, Chris Chinenye Emezue, Bonaventure

F. P. Dossou, Blessing Sibanda, Blessing Bassey, Ayodele Olabiyi, Arshath Ramkilowan, Alp Öktem, Adewale Akinfaderin, and Abdallah Bashir. 2020. Participatory research for low-resourced machine translation: A case study in African languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2144–2160, Online. Association for Computational Linguistics.

Alexander Quinn Nichol and Prafulla Dhariwal. 2021. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171.

Kelechi Ogueji, Yuxin Zhu, and Jimmy Lin. 2021. Small data? No problem! Exploring the viability of pretrained multilingual language models for low-resourced languages. In *Proceedings of the 1st Workshop on Multilingual Representation Learning*, pages 116–126. Association for Computational Linguistics.

Krishna Pillutla, Swabha Swayamdipta, Rowan Zellers, John Thickstun, Sean Welleck, Yejin Choi, and Zaid Harchaoui. 2021. MAUVE: Measuring the gap between neural text and human text using divergence frontiers. In *Advances in Neural Information Processing Systems*, volume 34, pages 4816–4828.

Maja Popović. 2015. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395. Association for Computational Linguistics.

Sello Ralethe. 2020. Adaptation of deep bidirectional transformers for Afrikaans language. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 2475–2478. European Language Resources Association.

Ehud Reiter. 2018. A structured review of the validity of BLEU. *Computational Linguistics*, 44(3):393–401.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.

Phillip Rust, Jonas Pfeiffer, Ivan Vulić, Sebastian Ruder, and Iryna Gurevych. 2021. How good is your tokenizer? On the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3118–3135. Association for Computational Linguistics.

Tim Salimans and Jonathan Ho. 2022. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations*.

Oleh Shliazhko, Alena Fenogenova, Maria Tikhonova, Anastasia Kozlova, Vladislav Mikhailov, and Tatiana Shavrina. 2024. mGPT: Few-shot learners go multilingual. *Transactions of the Association for Computational Linguistics*, 12:58–79.

Robin Strudel, Corentin Tallec, Florent Altché, Yilun Du, Yaroslav Ganin, Arthur Mensch, Will Grathwohl, Nikolay Savinov, Sander Dieleman, Laurent Sifre, and Rémi Leblond. 2022. Self-conditioned embedding diffusion for text generation. *arXiv preprint arXiv:2211.04236v1*.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics.

Yizhe Zhang, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Joshua Susskind, and Navdeep Jaitly. 2023. PLANNER: Generating diversified paragraph via latent language diffusion model. In *Advances in Neural Information Processing Systems*, volume 36, pages 80178–80190.

## A Model and estimators as implemented

**Autoencoder.** The mT5 encoder maps a passage of at most 128 tokens to hidden states, which the Perceiver compresses to $u \in \mathbb{R}^{32\times64}$. Each slot is rescaled to norm eight, $x_j = 8\,u_j/\|u_j\|$, so that $x$ has unit variance per dimension on average, and

a reconstruction map turns $x$ into the decoder's cross-attention input. Training minimises token-weighted teacher-forced cross-entropy over the non-padding target tokens $y_1, \ldots, y_N$ of a batch, $\mathcal{L}_{\mathrm{AE}} = -\frac{1}{N}\sum_n \log p_\phi(y_n \mid y_{<n}, x)$.

**Forward process.** With $t \sim \mathcal{U}(0,1)$ and $\epsilon \sim \mathcal{N}(0, I)$, the noised latent is

$$z_t = \sqrt{\gamma(t)}\, x + \sqrt{1-\gamma(t)}\, \epsilon, \qquad (1)$$

with $\gamma(t) = \cos^2(\pi t/2)$, where $\gamma$ is the cumulative signal power ($\bar{\alpha}$ in DDPM notation), clipped so that $\log \frac{\gamma}{1-\gamma} \in [-15, 15]$. Latents are not shifted or scaled further.

**Objective.** The denoiser $f_\theta(z_t, t, c)$ predicts the velocity $v = \sqrt{\gamma}\,\epsilon - \sqrt{1-\gamma}\, x$, with unweighted squared error averaged over the $32 \times 64$ entries and the batch:

$$\mathcal{L} = \mathbb{E}_{x,t,\epsilon}\Big[\tfrac{1}{2048}\big\| f_\theta(z_t, t, c) - v \big\|_2^2\Big]. \qquad (2)$$

Predictions convert as $\hat{x}_0 = \sqrt{\gamma}\, z_t - \sqrt{1-\gamma}\, \hat{v}$ and $\hat{\epsilon} = \sqrt{1-\gamma}\, z_t + \sqrt{\gamma}\, \hat{v}$.

**Self-conditioning.** With probability 0.5 the input $c$ is a first-pass estimate, $c = \mathrm{sg}[\hat{x}_0(z_t, t, \varnothing)]$ with each slot rescaled to norm eight; otherwise $c = \varnothing$, a learned vector repeated over slots. During sampling, $c$ is the previous step's $\hat{x}_0$ *without* that rescaling, so training and sampling present self-conditioning inputs of different norm. Its effect is examined in §8.

**Sampler.** Sampling starts from $z_1 \sim \mathcal{N}(0, I)$ and takes 250 equal steps $t \to s$ with $s = t - 1/250$. With $\rho = \gamma(t)/\gamma(s)$ and fresh $\xi \sim \mathcal{N}(0, I)$,

$$z_s = \frac{1}{\sqrt{\rho}}\Big(z_t - \frac{1-\rho}{\sqrt{1-\gamma(t)}}\,\hat{\epsilon}\Big) + \sqrt{1-\rho}\,\xi, \quad (3)$$

and the last step returns $\hat{x}_0$, which is decoded without renormalisation. Checkpoints use exponential moving averages of the weights (decay 0.9999) and are selected on $\mathcal{L}$ estimated on 2,048 validation latents, three times with fixed seeds.

**Scoring.** Each output $i$ is tokenised by the scorer into $n_i$ tokens and scored after a start token, each token once, in 512-token windows with stride 256:

$$\mathrm{PPL} = \exp\Big(\frac{\sum_i \sum_{j=1}^{n_i} -\log p(w_{ij} \mid w_{i,<j})}{\sum_i n_i}\Big). \qquad (4)$$

No end token is scored. For length-standardised perplexity, $\ell_b$ is the same token-weighted loss within word-length band $b$, and $\mathrm{PPL}_{\mathrm{std}} = \exp\big(\sum_b w_b\, \ell_b\big)$ with $w = (0.140, 0.223, 0.349, 0.288)$, the band shares of 1,000 human validation passages; it is reported only if every band holds at least 20 outputs. The large-minus-small permutation test is two-sided over all 252 divisions of the ten run means.

## B A second language: Swahili

To test whether the checks matter beyond Afrikaans, we applied them, without retraining, to an earlier Swahili run of the pipeline that had been evaluated only by denoising loss and perplexity. Its corpus, 738 supplied text files (23.6 million words) without recoverable document boundaries, was split by file into windows of at most 48 words (437,326 training, 29,761 validation, 25,537 test), and the files share some text across splits. The run used the mT5 tokeniser, two seeds per size and equal wall-clock budgets (about 172,000 updates small, 53,000 large), with no AR baseline. Its test mGPT perplexity was 19.5 and 19.6 (small) and 20.6 and 20.8 (large), against 13.8 for human text. The checks were approved in advance and used validation data only. Reconstruction of 200 passages reached mean chrF 74.8 (95% CI [72.9, 76.5]) against 24.3 with shuffled latents, but no exact matches, and numeral recall was 0.54. Random latents decoded by the same decoder scored clearly worse (perplexity 135) than diffusion samples. GlotLID labelled all diffusion samples and references as Swahili, against 57% of random-latent outputs. Diffusion outputs shared 10–11% of their word 4-grams with the training split, against 28% for references, which the supplied splits' cross-split overlap inflates. All readings are automatic. Fluent-scoring samples again coexist with an autoencoder that reconstructs no passage exactly, but Swahili random latents score far worse than diffusion samples, as on the retrained Afrikaans autoencoder.

## C Evaluation sets

Table 6 lists every set behind the reported numbers. Table 7 gives the values plotted in Figure 1.

| Set | Size | Selection | Used for | First used |
|---|---|---|---|---|
| Training split | 47,767 passages, 2,005 docs | Document split, seed 42 | Training; overlap and copy checks | – |
| Audit passages | 81 passages, 64 docs | Validation docs 0–63 of seed-91427 shuffle | Reconstruction, interventions | 22 Oct. |
| Selection panel | 663 passages | Same 64 documents | Screen: early stopping, ranking | 27 Oct. |
| Comparison panel | 194 passages | Validation docs 128–250 | Screen contrasts; train/held-out check | 27 Oct. |
| Reserve | 103 passages | Validation docs 64–127 | Screen winner vs original, once | 31 Oct. |
| Human validation pool | 4,825 passages | Non-reserve validation, $\leq$128 tokens | Language, dictionary, overlap | 31 Oct. |
| Human scored subset | 811 passages, 58 docs | 1,000 scored validation rows minus reserve docs | Scorers; reconstruction of human text | 30 Oct. |
| Diffusion validation | $5 \times 1{,}000$ per size | Sampling seeds 30000–30999 | Controls, language, scorers | Before 23 Oct. |
| 256-row subset | 256 per run | First 256 seeds | AR ratios | 22 Oct. |
| AR samples | 256 | Nucleus sampling | AR ratios, language | 22 Oct. |
| Latent controls | 1,000 per control | Seeds 30000+, 40000+, 50000+ | Decoder-only controls | 30 Oct. |
| Test banks | $5 \times 1{,}000$ per size; 1,000 human | Seeds 120000–120999 | Test PPL, Figure 1 | 19 Sept. (AE), 9 Oct. (PPL) |
| Extra-data monitor | 681 passages | 5% of extra-data documents | Fresh reconstruction check (original; A and D on 4 Nov.) | 1 Nov. |
| Audit seeds | 100 per checkpoint | Seeds 70000–70099 | Self-conditioning audit | 4 Nov. |
| AR test samples | 1,000 | Seeds 120000–120999 | AR test ratios | 5 Nov. |

Table 6: Evaluation sets. "First used" is when results on the set were first seen; the test banks were exposed before the models were frozen on 23 October, including an unblinded AI rating of 20 test outputs per configuration on 9 October (not retained).

| Words | 1–10 | 11–20 | 21–35 | 36+ |
|---|---|---|---|---|
| Human | 240.8 | 54.1 | 32.6 | 25.9 |
| passages | 157 | 212 | 300 | 331 |
| Small, mean | 332.1 | 89.4 | 53.6 | 40.2 |
| range | 294.7–386.9 | 82.4–94.5 | 52.2–55.0 | 39.1–40.8 |
| outputs | 797 | 851 | 1,419 | 1,933 |
| Large, mean | 213.1 | 69.3 | 47.4 | 38.1 |
| range | 188.0–246.1 | 63.7–73.3 | 46.4–48.7 | 37.6–39.3 |
| outputs | 739 | 1,262 | 1,579 | 1,420 |
| AR | 272.1 | 77.8 | 64.3 | 57.4 |
| outputs | 315 | 352 | 219 | 114 |

Table 7: Test mGPT perplexity by output length (Figure 1). Diffusion: mean and range of five run values; outputs summed over runs. AR: 1,000 samples. The range is not a confidence interval.